



## Enrichment Considerations

*Enrichment definition: Enrichment is a dynamic process for enhancing animal environments within the context of the animals' behavioral biology and natural history. Environmental changes are made with the goal of increasing the animal's behavioral choices and drawing out their species-appropriate behaviors, thus enhancing animal welfare. (Aza/Bag)*

Behavioral enrichment is an integral part of daily management of animals in human care, which enhances the lives of animals. A good enrichment strategy has several guiding principles:

1. The strategy should be proactive, not reactive with safety in mind
2. The strategy should allow all animals to have some control and choice over their environment
3. The strategy should be based upon the animals' biological, social, cognitive needs
4. The strategy should encourage species-appropriate behaviors

When designing, installing and implementing enrichment, the importance of following established safety protocols cannot be understated. To avoid potential animal enrichment hazards, the following guidelines should be utilized:

### General safety considerations

- If there are more than one animal within the habitat, provide sufficient items to help prevent fighting between the animals. It is recommended to provide at least one item per animal in the exhibit
- Non-toxic materials must be used if animals will be in contact with the item.
- Natural items are preferred when possible to blend in with the habitats. Non-natural material such as plastics or nylon may be used but may be camouflaged for habitat use
- When constructing an item, ensure all corners are rounded off and objects are sanded if needed so there are no splinters or rough edges. If hardware is used, it must be stainless steel and locking to ensure it cannot be removed by animals. If possible, use screws and non-toxic glue instead of nails.
- Be aware of any pinch points or entrapment areas. Openings such as chains must ensure no part of the animal can be entrapped. This will be unique to each animal species. Use of PVC or other coverings can ensure chains are used safely.



- Care must be used regarding possible ingestion. All items must be inspected for small pieces and instabilities that would allow an animal to chew or remove pieces for ingestion.
- Use of staples, plastic tape, or bags with plastic liners are not allowed
- Do not use any stains or treated wood unless approved

Examples of questions to ask when adding new enrichment items or strategies:

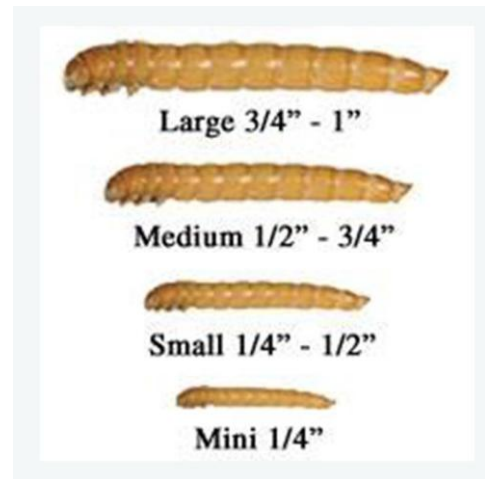
1. Does the item pose any danger to the animal including being injured, caught up, or entrapped?
2. Can it result in one animal injuring another during use?
3. Can it fall on another animal in the enclosure?
4. Can it cause damage/destroy the habitat?
5. Can the animal throw the item outside the habitat at other animals or guests?
6. Will the item cause displacement or fighting among the animals on exhibit?
7. If the enrichment includes food, is it a reasonable amount/size?
8. Can the animal choke on the item?
9. Will the item cause high stress levels for the animal for an unhealthy period?
10. Can the animal use the item to escape?
11. Do the benefits outweigh the costs or vice versa?
12. Are there any unique aspects regarding the species, individuals, or habitat, which will provide unique challenges?

## **Enrichment Considerations- Mealworm-Darkling beetle hide:**

The OKC Zoo cares for a wide variety of animals including invertebrates. These insects are vital for healthy environments and serve as ambassadors for our guests to learn and appreciate lesser appreciated animals in the world. They also provide a vital food source to some of our larger species. The care and welfare we provide these animals directly translates to their life as well as the nutrition they provide other animals. We strive to ensure all beings in our care have the best life possible so ensuring our insects have fulfilling habitats is equally as important as our more charismatic species.

The Darkling beetle is one of our more numerous species we care for. Darkling beetles occupy ecological niches in mainly deserts and forests as plant scavengers. Most species are generalistic omnivores, and feed on decaying leaves, rotting wood, fresh plant matter, dead insects, and fungi as larvae and adults. The larvae, known as mealworms or false wireworms, are usually fossorial, heavily sclerotized and nocturnal. They are an important resource for certain invertebrates and small mammals. Adults of many species have chemical defenses and are relatively protected against predators. Adults of most species, except grain pests, have slow metabolisms, and live long lives compared to other insects, ranging from approximately six months to two years. Some species live in intensely dry deserts such as the Namib, and have evolved adaptations by which they collect droplets of fog that deposit on their elytra. As the droplets accumulate the water drains down the beetles' backs to their mouthparts, where they swallow it.





#### Enrichment Ideas:

One of the most important behavioral needs of this species is hiding/denning locations. The ability to hide allows for breeding and egg laying as well as lowering stress levels. Caretakers are requesting options to create more complexity as well as different designs so the insects can exercise preference in where they hide. Due to the size of the enclosures, hides need to fit the following: 8-10" long at max for the small bins and 12-15" long for the big bins. Both should be probably 4" wide at max and 3-4" tall at max.

The safest materials would be acrylic or a sturdy plastic (they will chew thinner plastics). Wood cannot be used due to the animals eating it. For both materials there will need to be some type of grip/rough texture for them to walk on due to it being a slick texture. This could be anything like a stippling effect or lines that are carved into the material so their little feet can get into them. All forms should also have ventilation. Both the mealworms and beetles love to dig pile into things which can become deadly. Vertical designs such as tubes, steps, etc. with textured sides are great so they can take advantage of elevation for temperature control. Again, ventilation is needed as well as multiple entrances and exits. They don't do reverse well. Hides may be set on the surface or partially buried depending on the design.

Animals currently have cardboard tubes and cartons however; these cannot be sanitized resulting in us throwing out a lot of material. Additionally, they do not allow for complexity such as shape and different exits/entrances. They also require donations which can fluctuate throughout the year.



**Habitat:**

The habitat consists of. **Measurements are as follows: Small bin type is 12"x6" and the large bin type is 18"x12".**

**Small Bins:**



Large bins:

